

ORIGINAL ARTICLE

Efficacy of ceramides and niacinamide-containing moisturizer versus hydrophilic cream in combination with topical anti-acne treatment in mild to moderate acne vulgaris: A split face, double-blinded, randomized controlled trial

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Abstract

Introduction: Topical therapy is the mainstay treatment of acne, and topical retinoids such as tretinoin, tazarotene, and adapalene are recommended as the first-line therapy for mild to moderate acne. However, the cutaneous irritations may occur, and the dermocosmetics are recommended to prevent side effects of anti-acne drugs and adhere to treatment. Thus, this study aims to compare the efficacy and tolerability of ceramides and niacinamide-containing moisturizer (CCM) versus hydrophilic cream in combination with topical anti-acne treatment in mild to moderate acne vulgaris.

Methods: This was an 8-week, randomized, double-blinded, split face study in 40 patients assigned for topical anti-acne medications (5% benzoyl peroxide and 0.1% adapalene gel), then randomly applied CCM or hydrophilic cream. All patients were followed at week 0, 2, 4, and 8 for acne improvement, adverse reactions, biometric, and biophysical evaluation.

Results: CCM could significantly improve the non-inflammatory, inflammatory, and total acne lesions compared with hydrophilic cream after week 8 of treatment. Interestingly, there was an improvement of global worst score, hemoglobin index, melanin index, TEWL, skin hydration, sebum production, and skin surface pH, with no statistically significant differences between the two treatments. No serious side effects from clinical application of CCM and hydrophilic cream in mild to moderate acne vulgaris patients.

Conclusion: Ceramide and niacinamide-containing moisturizer in combination with anti-acne medication can significantly improve acne lesions and decrease cutaneous irritations toward a satisfactory treatment outcome of mild to moderate acne vulgaris.

KEYWORDS

acne treatment, ceramides, moisturizer, niacinamide, randomized controlled trial

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1 | INTRODUCTION

Acne is a chronic inflammation of the pilosebaceous unit with a high prevalence of disease during adolescence. Topical therapy is the mainstay treatment of acne and topical retinoids such as tretinoin, tazarotene and adapalene are recommended as the first-line therapy for mild to moderate acne, according to the international guideline by the Global Alliance.¹

Although irritation side effects from adapalene are less common than other formulations of topical retinoids,^{2–4} cutaneous irritations e.g., erythema, desquamation, burning, and pruritus may occur. Various attempts have been designed to minimize or reduce these side effects. Non-comedogenic moisturizers are frequently added as an adjunctive therapy to relieve cutaneous irritations from acne medications.⁵ Moreover, some moisturizers also contain ingredients with anti-inflammatory properties which may improve acne.⁶ Recent advances in science and technology have showed the synergistic role of dermo cosmetics in the treatment of acne vulgaris to prevent side effects of anti-acne drugs and adhere to treatment.⁷ Dermatologists often suggests using moisturizers for a wide range of skin problems initiated by acne treatment, such as ceramides containing moisturizer (CCM), to simulate skin moisturization.⁸ Moreover, topical niacinamide application can also help improve acne vulgaris due to its combination of anti-inflammatory action and reduction of sebum production.⁶ Nonetheless, there have been limited studies on the tolerability and efficacy of those moisturizers for acne vulgaris treatment. Thus, this study aims to compare the efficacy and tolerability of CCM versus hydrophilic cream in combination with topical anti-acne treatment in mild to moderate acne vulgaris.

2 | MATERIALS AND METHODS

2.1 | Study design

This was a split face, double-blinded, randomized controlled trial. The computer-generated randomization was applied for CCM versus hydrophilic cream in combination with topical anti-acne treatment packaging. The randomization was blinded to dermatologists and patients. Informed consent was obtained from all participants in the study and all procedures were approved by the Human Ethics Committee of Thammasat University (MTU-EC-OO-6-210/63).

TABLE 1 Summary of ingredients in CCM and hydrophilic cream.

	Occlusive	Humectant	Emollient	Preservative	Active Ingredients
CCM	-Dimethicone -Caprylic/Capric triglyceride -Cholesterol -Cetyl alcohol	-Glycerin -Sodium hyaluronate	-Cetearyl alcohol -Cetyl alcohol -Propylparaben	Phenoxyethanol	-Ceramide NP -Ceramide AP -Ceramide EOP -Phytosphingosine -Niacinamide
Hydrophilic cream	-Cetyl alcohol -Stearic acid	Propylene glycol	-Cetyl alcohol -Propylparaben	Propylparaben	

[Correction added on 13 February 2024 after first online publication: Table 1 has been updated in this version.]

2.2 | Study population

The study population were Thai males and females, aged 18 to 40years, with mild to moderate acne vulgaris from Global Acne Grading System (GAGS) at Benchakitti Park Hospital, Bangkok, Thailand.

2.3 | Sample size calculation

The sample size was calculated based on the proposed adjunctive usage of a non-comedogenic moisturizer with adapalene gel 0.1% to improve local tolerance,⁹ 5% alpha and 20% beta errors. This calculation yielded an initial sample size of 34, accounting for a 20% dropout rate. The final sample size was determined to be 40.

2.4 | Selection criteria

The inclusion criteria were as follows: healthy males and females aged 18–40years, Fitzpatrick skin Types III to V, capable of reading and signing the informed consent, individuals with mild and moderate acne vulgaris according to the Global Acne Grading System score (GAGS), non-pregnant and non-lactating women, no use of topical and systemic acne treatments within 4weeks prior to enrollment, no previous laser and light therapy, topical corticosteroids, dermabrasion, or peeling within the past 4weeks, no use of systemic retinoids, oral contraceptive pills, or hormonal therapy within the past 6months, absence of severe acne, no history of photosensitivity, no systemic diseases or other facial skin disorders apart from acne vulgaris, and no history of allergies to the study products.

2.5 | Procedures and measurements

Patients were assigned for a twice daily treatment of 5% benzoyl peroxide (BP) on the entire face in the morning after facial washing (using gentle facial soap), then randomly applied ceramide-containing moisturizer (CCM, CeraVe PM Facial Moisturizing Lotion, Valeant Pharmaceuticals North America LLC.) on one side of the face and hydrophilic cream on the opposite side for 8weeks. Also, 0.1% adapalene gel was given as a night treatment on the entire face with both moisturizers after facial washing. All patients

were followed at week 0, 2, 4, and 8 for adverse reactions, acne improvement, as well as treatment tolerability and efficacy. The comparison of ingredient lists between the two moisturizers, clarifying the differences between CCM and the hydrophilic cream, is presented in [Table 1](#).

Treatment outcomes were measured by Global Worst Score, GAGS, Acne lesion counts (non-inflammatory, inflammatory and total acne lesions), Transepidermal water loss (TEWL) by Tewameter®, Skin hydration by Corneometer®, Skin surface pH by Skin pH-Meter PH 905®, Sebum production by Sebumeter® (SM 815-Courage-Khazaka Electronic, Köln®), Hemoglobin (redness), melanin index (hyperpigmentation) and skin texture by biometric camera (Antera 3D; Miravex Co., Ltd, Dublin, Ireland).

3 | RESULTS

3.1 | Demographic data

A total of 40 patients with mild to moderate acne vulgaris were enrolled in the study of 8 weeks. Most of them were female (70%). The mean age was 30.48 ± 7.44 year (range 18–40 years) and the mean duration of acne was 10.73 ± 6.80 year (range 1–28 years) ([Table S1](#)).

3.2 | Acne lesion counts ([Figure 1](#) and [Table S2](#))

Acne lesion counts were recorded on both sides of the face during follow-up visits at weeks 0, 2, 4, and 8. When discussing “improvements” in the lesion count in the study, it refers to a reduction in the number of acne lesions.

All of the enrolled patients were taken photograph at straight face, 45 degrees left/right and 90 degree left/right, for the evaluation of non-inflammatory, inflammatory, and total acne by acne lesion counts in each follow-up visit by the blinded investigator. Clinical visit at week 0, 2, 4, and 8 as shown in [Figure 1](#).

3.3 | Non-inflammatory acne counts

On the hydrophilic side, there was a statistically significant improvement on the non-inflammatory lesions compared with the baseline and at week 2 of treatment (20% improvement, p -value < 0.001). At week 4 of treatment, there was an improvement for 40% from the baseline (p -value < 0.001). Then, the improvement was 68.33% at week 8 of treatment from the baseline (p -value < 0.001).

On the CCM side, a statistically significant improvement was observed at week 2 of treatment (30% improvement, p -value < 0.001). At week 4 of treatment, there was an improvement of non-inflammatory lesions for 50% from the baseline (p -value < 0.001). Then, the improvement was 80% at week 8 of treatment from the baseline (p -value < 0.001).

When comparing between both groups, there was a significant difference at week 8 of treatment. The improvement was 68.33% on the hydrophilic side and 80% on the CCM side (p -value = 0.022).

3.4 | Inflammatory acne counts

There was no statistically significant improvement for the inflammatory lesions on the hydrophilic side until the end of the study when compared with the baseline.

A statistically significant improvement was observed on the CCM side at week 4 of treatment (100% improvement, p -value = 0.003). At week 8 of treatment, the improvement was also 100% from the baseline (p -value = 0.004).

When comparing between the two groups, there was a significant difference at week 2 (p -value = 0.027) until week 8 (p -value = 0.002) of treatment.

3.5 | Total acne counts

There was a statistically significant improvement on the hydrophilic side when compared with the baseline at week 2 of treatment 20% improvement (p -value < 0.001). At week 4 of treatment, the improvement was 40% from the baseline (p -value < 0.001). At week 8 of treatment, the improvement was 63.45% from the baseline (p -value < 0.001).

On the CCM group, a statistically significant improvement was observed at week 2 of treatment (28.64% improvement, p -value < 0.001). At week 4 of treatment, the improvement was 51.09% from the baseline (p -value < 0.001). At week 8 of treatment, the improvement was 77.5% from the baseline (p -value < 0.001).

When comparing both groups, there was a significant difference at week 8 of treatment. The improvement was 63.45% on the hydrophilic side and 77.5% on the CCM side (p -value = 0.008).

3.6 | Biometric skin parameter evaluation ([Table S3](#))

The patients were photographed by digital camera in 5 angles: straight, 45 and 90 degrees on the left, 45 and 90 degrees on the right. They were also taken Antera 3D photograph® on both cheeks during every follow-up visit to evaluate hemoglobin (redness), melanin (post inflammatory hyperpigmentation, PIH), texture, roughness (scale) and wrinkle.

3.7 | Hemoglobin index

The hemoglobin index showed erythema from inflammation of the skin. Clinical visits were scheduled at baseline, week 2, week 4, and

FIGURE 1 Evaluation of hydrophilic cream versus CCM with topical anti-acne treatment in mild to moderate acne vulgaris patients using the split face study using standardized photography, hemoglobin index and melanin index at baseline, week 2, week 4, and week 8.



week 8. When comparing between both groups, no statistically significant differences were observed (p -value=0.767).

3.8 | Melanin index

The melanin index showed hyperpigmentation from inflammation of the skin. Clinical visits were scheduled at baseline, week 2, week 4, and week 8. When comparing between the two groups, no statistically significant differences were observed (p -value=0.881).

3.9 | Biophysical skin parameter evaluation (Figure 2 and Table S3)

There was the biophysical skin parameter evaluation for TEWL, skin hydration, sebum production, and skin surface pH measurement at baseline, week 2, week 4, and week 8 of treatment.

3.9.1 | Transepidermal water loss (TEWL)

The baseline on both hydrophilic side and CCM side did not show any statistically significant difference (19.55 ± 8.80 and 18.29 ± 7.77 , respectively). When comparing at baseline, the TEWL measurement at week 8 of treatment on hydrophilic side and CCM side remained higher. There were no statistically significant differences between

the two groups at baseline (p -value=0.501) until the end of the study (p -value=0.563).

3.9.2 | Skin hydration

The skin hydration of both sides was similar (Figure 2B). There were no statistically significant differences between the two groups at baseline (p -value=0.551), week 2 (p -value=0.485), week 4 (p -value=0.076), and until the end of the study (p -value=0.711).

3.9.3 | Sebum production

The baseline of sebum production for both hydrophilic side and CCM side did not show any statistically significant differences (23.23 ± 16.62 and 23.89 ± 16.96 , respectively) (p -value=0.829). There were no statistically significant differences between the two groups at baseline (p -value=0.829) until the end of the study (p -value=0.743).

3.9.4 | Skin surface pH

The baseline of skin surface pH for both hydrophilic side and CCM side showed no statistically significant difference (5.61 ± 0.40 and 5.61 ± 0.38 , respectively) (p -value=0.946). There were no

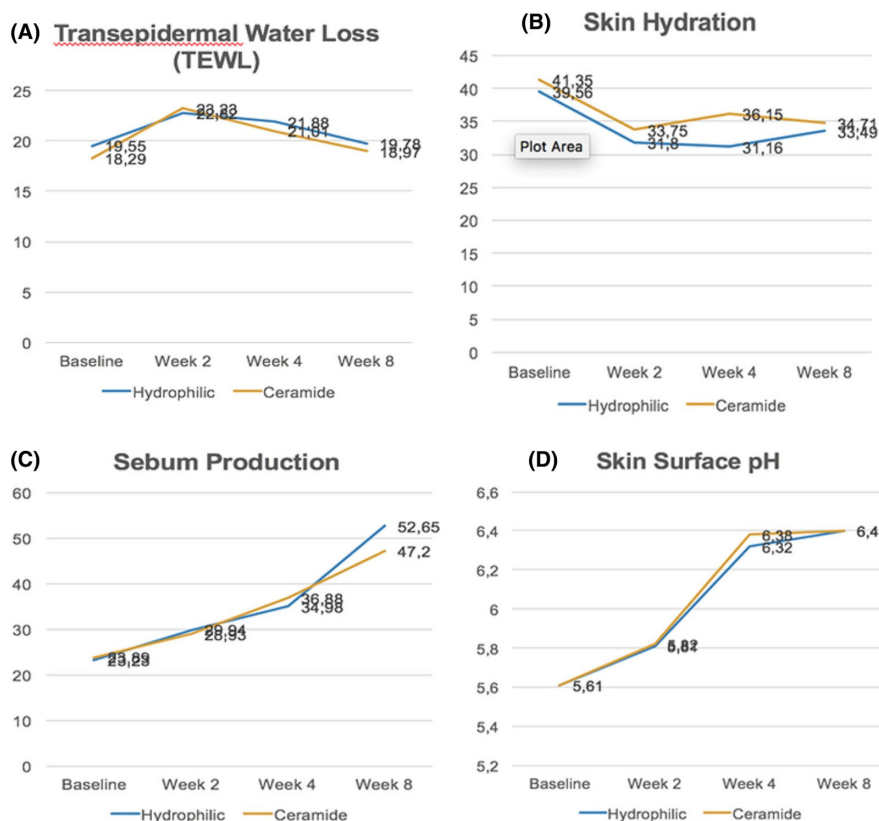


FIGURE 2 Measurement of TEWL (A), skin hydration (B), sebum production (C) and skin surface pH (D) of hydrophilic and CCM sides on the baseline, week 2, week 4, and week 8 of treatment, ($n=40$).

statistically significant differences between the two groups at baseline (p -value=0.946), week 2 (p -value=0.953), week 4 (p -value=0.996), and until the end of the study (p -value=0.945) (Figure 2D).

3.10 | Global Worst Score

Local tolerance was evaluated by using the score: 0-none; 1-mild; 2-moderate; 3-severe for erythema, scaling, dryness, stinging or burning, and pruritus on each half-face at each follow-up visit. The Global Worst Score was calculated as the mean of five worst scores.⁹

Most of the patients in all groups reported "none" or "mild" irritation for each parameter. However, there were no significant differences in all parameters among the two groups after 2–8 weeks of treatment in each parameter. (Table S4).

4 | DISCUSSION

The study was conducted over 8 weeks in 40 patients with mild to moderate acne vulgaris, in line with the duration of previous studies involving adjunctive moisturizer for acne treatment.^{9,10} Its objective was to assess the efficacy and skin tolerability of CCM when compared to a hydrophilic cream used in combination with topical anti-acne treatment.

In this study, we found that most of the patients in all groups were reported clinical evaluation as none or mild irritation for each parameter. For erythema, more than 65% of the patients of all groups reported none or mild irritation and desquamation. Also, more than 90% of the patients of all groups had none or mild irritation. Whereas, more than 87.5% of the patients of all groups revealed none or mild irritation for dryness, burning or stinging sensation and pruritus. The results yielded that adjunctive usage of moisturizers and topical acne medications could help reduce skin irritation initiated by anti-acne treatment. When compared between the two groups, there were no significant differences for each parameter between both groups during 8 weeks of treatment.

To determine the efficacy, acne lesion counts were evaluated at baseline, week 2, week 4, and week 8 of treatment. Our study showed that non-inflammatory and total acne lesions decreased significantly in all groups at week 2 of treatment compared with the baseline. When comparing between both groups, the CCM significantly provided better improvement on non-inflammatory and total acne lesions compared with hydrophilic cream after week 8 of treatment. This favorable outcome might be due to the benefit of ceramides replacement in acne vulgaris patients. According to previous studies, the skin of patients with acne vulgaris was associated with ceramides deficiency, which correlated directly to greater acne vulgaris severity. Skin barrier permeability was theorized to contribute to follicular hyperkeratosis and formation of comedones.¹¹ Linoleate in ceramide was decreased among acne vulgaris patients.¹²

Recent research has underscored the significance of ceramides in addressing skin barrier deficiencies, which is commonly found in acne patients, resulting in dryness and irritation.¹³ Applying external ceramides within moisturizers has demonstrated positive outcomes in reducing lesions, enhancing ceramide composition, and fortifying the skin barrier.¹⁴ This could underscore the potential of ceramide-containing products as active cosmetic ingredients for improving acne and skin condition.¹⁵ These products, including ceramides, phytosphingosine, dimethicone, glycerin, and hyaluronic acid, facilitate epidermal differentiation, maintain skin hydration, and establish an optimal healing environment. In particular, ceramides play a pivotal role in barrier repair.¹⁶ The in vitro and in vivo studies showed a strong antimicrobial effectiveness of skin identical phytosphingosine produced by ceramides. For both in vitro and in vivo, the phytosphingosine-containing product showed excellent clinical results in the context of skin care in acne, based on both anti-inflammatory and antimicrobial activity. These results demonstrate the potential of ceramide-containing products to enhance or complement existing acne therapies which act as active cosmetic ingredients.¹⁷

Another active ingredient, niacinamide, a form of vitamin B3, has been substantiated by various studies to be effective in treating acne due to its multifaceted role in reducing sebum production,¹⁸ inflammation,¹⁹ and bacterial growth²⁰ while enhancing the skin barrier,²¹ reducing hyperpigmentation,²² and improved the appearance of acne scars and discoloration.²³ Niacinamide supports mechanisms of cellular energy and DNA repair which contribute to its potential benefits in various skin conditions, including acne.¹⁹

In terms of the results concerning inflammatory acne lesions, our study revealed that there was no flare-up of inflammatory lesions in the CCM group at week 4 of treatment, in contrast to the hydrophilic group. When comparing the two groups, it became evident that the CCM group exhibited significantly better improvement in inflammatory acne lesions from week 2 of treatment until the end of the study. While the exact mechanism of this superiority remains unclear, it is possible that the ceramide properties, the anti-inflammatory effects, and reduced sebum production attributed to niacinamide may contribute to these favorable outcomes. However, further research is needed to elucidate the precise mechanisms involved. Topical retinoids often trigger flare-ups of inflammatory acne lesions during the initial weeks of treatment. The anti-inflammatory properties of niacinamide may prove beneficial in minimizing flare-ups, thus promoting greater patient compliance.⁷

In this study, we also evaluated the post-acne erythema and PIH in both groups by biometric camera to access the hemoglobin index and melanin index reduction at baseline, week 2, week 4, and week 8 of treatment. Acne-related erythema was caused by angiogenesis after inflammatory reaction and niacinamide had anti-inflammatory properties which could improve the function of the skin barrier for less irritation and reddening. PIH was also a particularly common sequela after acne vulgaris and niacinamide had the ability to reduce melanosome transfer from melanocytes to surrounding keratinocytes, as well as improve the PIH.^{7,24} Our study demonstrated

slightly decreased hemoglobin and melanin index, but no statistically significant differences were observed between both groups.

Moreover, we also evaluated the biophysical parameter, including TEWL, skin hydration, sebum production, and skin surface pH. Topical anti-acne medications are known to cause skin irritation, while controverting stratum corneum barrier function during treatment. The expected results were to increase TEWL with decreased skin hydration.²⁵ The using of moisturizers in combination with topical acne medications should have beneficial effects on managing the TEWL and skin hydration. However, our study found that the TEWL was still slightly increased and skin hydration was slightly decreased in all groups at baseline until week 8 of treatment, with no significant differences between the two groups.

Excess sebum production is one of the main pathogeneses of acne.²⁶ The decreased sebum production effect of niacinamide in the study moisturizer should have beneficial effects on reducing the sebum production.⁷ However, the sebum production was still increased in all groups of treatments at baseline until week 8, with no significant differences between the two groups.

The increased facial skin pH in patients with acne at basal conditions mirrors a chronic state of stratum corneum instability which could predispose individuals to acne occurrence and/or recurrences.²⁷ Ceramides could be one of the essential lipid to restore barrier function and maintain the buffer system of the skin, resulting in regulation of physiological environment for skin repair itself and skin surface pH, leading to acidic pH.²⁸ However, our study showed that the skin surface pH was still increased in all groups at baseline until week 8 of treatment, with no significant differences between the two groups. There were several factors contributing to the deviation result of excessive sebum production and skin pH, including both external and internal factors such as the age of participants, illness, oral medications, time and seasons of this study, and others.

4.1 | Limitations

One of the limitations of this study was face mask confounder, which might influence the outcome of the study because of the irritation from mask friction or allergy, while the humidity and heat under the mask might cause the skin capillaries to dilate. The sample size was quite small which might make it difficult to determine if a particular outcome was an actual finding. The research was also done during the COVID-19 pandemic which might influence the participant's concern to the pandemic situation rather than using the medications properly.

5 | CONCLUSION

Ceramide and niacinamide-containing moisturizer can significantly improve the non-inflammatory, inflammatory, and total acne lesions compared with hydrophilic cream after week 8 of treatment. Interestingly, there are no statistically significant differences

between CCM and hydrophilic cream in the global worst score, hemoglobin index, melanin index, TEWL, skin hydration, sebum production, and skin surface pH during 8 weeks of the study. No serious side effects are observed from clinical application of CCM and hydrophilic cream in mild to moderate acne vulgaris patients.

Nonetheless, further studies of CCM moisturizer and other active ingredients in moisturizer as an adjunctive treatment of acne vulgaris would be suggested in larger sample size to verify the effectiveness in acne vulgaris treatment.

FUNDING INFORMATION

No funding was provided for this article.

CONFLICT OF INTEREST STATEMENT

The authors report no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ETHICS STATEMENT

All procedures performed in studies involving human participants were in accordance with the ethical standards of the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Informed consent was obtained from all participants in the study, and all procedures were approved by the Human Ethics Committee of Thammasat University (MTU-EC-OO-6-210/63).

STUDY POPULATION

The study population were Thai males and females, aged 18 to 40 years, with mild to moderate acne vulgaris from Global Acne Grading System (GAGS) at Benchakitti Park Hospital, Bangkok, Thailand.

PATIENT CONSENT

The patient signed the informed consent form after understanding the nature of the trial.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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